ly229 at georgetown.edu & (215) 688-6404

Lu Yu, Ph.D.

Interests

Quant Research & Agentic AI on Economics

Education

Ph.D., Georgetown University, Economics, (Mentors: Prof. Toshihiko & Prof. James Angel) 2019 ~ 2025
B.S., University of Pittsburgh, Mathematics & Economics (**Honor, 4.0/4.0**) 2015 ~ 2018

Skills

Certificate: Nvidia Deep Learning, Investment Banking by J.P. Morgan

Program, Software Skills and Database:

- **Programming & Data**: Python, R, Stata, SQL, MATLAB, Excel, LaTeX
- **Quantitative Methods**: Equity derivatives pricing and validation, barrier options, multi-asset structures, Monte Carlo simulation, Black–Scholes, local and stochastic volatility, PDE methods, Greeks and sensitivity analysis, convergence testing, backtesting, P&L analysis, econometrics, time-series analysis, market microstructure, risk analysis, financial forecasting, portfolio analysis
- **AI & Cloud Tools**: PyTorch, LangGraph, Codex, Harbor, AWS (EC2, S3)

Work Experiences

Morgan Stanley Associate, Equity Valuation — Baltimore, MD, Aug 2025 ~ Present

- Develop AI-AGENT workflow for Risk Management. (Co-Pilot & CODEX CLI)
- Price equity derivatives, including barrier options and multi-asset structures.
- Validate pricing models under Black–Scholes and stochastic-volatility frameworks through Greek and sensitivity analysis, convergence testing, backtesting, and P&L analysis.
- Conduct risk modeling across AML, crypto, financial crime, and market risk domains.
- Ranked as a top performer, selected for **Super Tier 1** projects.
- Demonstrated strong execution delivering time-critical risk reports across global teams.

Georgetown University, Ph.D — Washington, DC 2019 ~ 2025

- **Instructor: Department of Finance**
  - OpenFinTech: An open source project to build customized DeepAgent AI Workflow
  - Design an AI Assistant for FINRA CRCP Grading (deepagent powered)
  - Conducted literature reviews, prepared talking points, maintained an accurate research database.
  - Performed quantitative data analysis, applying statistical methods to draw actionable conclusions.
- **Research Assistant: McDonough School of Business**
  - Deep Learning on financial data analysis. Modeling Currency Volatility by using RNN(LSTM).
  - Analyzed exchange-rate drivers, periodic behavior;
  - Applied Monte Carlo Simulation to evaluate financial risk and model uncertainty.
- **Teaching Assistant: Department of Economics**
  - Supported courses: corporate finance, financial econometrics, valuation & risk management.
  - Key topics: Regression analysis, capital structure, free cash flow valuation & derivatives markets.
  - Trained students to examine, manipulate, and present data in Excel & JMP.

Publications & Patents & Presentations

- PhD Thesis: ESSAYS ON INDEX OWNERSHIP AND MARKET MICROSTRUCTURE **Lu Yu**, Thesis (2026)
- From Clerks to Agentic-AI: How will Technology Change Labor Market in Finance? **Lu Yu**, Xiang Li, arXiv (2026)